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# Aaditya Tamrakar

*Software Engineer · Quant Developer · India, open to relocation*

Quant developer and software engineer in Python, Go and TypeScript. Currently building Greeks-based options strategies and MT5 trading systems as a freelancer; previously shipped 80+ order execution algorithms handling ₹200 crore at Algo Maker and production Go services, data pipelines and CI/CD at Viasat.

## Work Experience

**Independent** *Quant Researcher & Developer* Remote · Jan 2026 – Present

- Built 50+ MT5 Expert Advisors for gold CFDs; the 5 to 6 in production use recovery-factor based risk and run at 80%+ trade accuracy for freelance clients.
- Designed systematic intraday Nifty straddle strategies: Supertrend and other indicator signals on the ATM straddle premium series, Greeks-based position management, automated entry and exit via broker APIs.
- Backtested 300+ strategy variants across 20+ technical indicators on US and Indian index and stock options history; monitor the live ones on Indian index options daily.

**Viasat Inc.** *Software Engineer II* Remote, US · May 2023 – Dec 2025 · Intern, Summer 2022

- Built a Go microservice that monitors servers deployed on aircraft, with the core platform team.
- Took a production API endpoint serving 2 million requests a day from a 50% failure rate to 100% success; owned Snowflake schema changes and Airflow DAGs for in-flight connectivity data.
- Owned product deployments: push-triggered GitHub Actions CI/CD with Python build and deploy scripts, CloudFormation stacks and AWS Secrets Manager, shipping to Kubernetes end to end; consolidated shared code into a common library with 100% unit test coverage.

**Algo Maker LLP** *Full Stack Developer* Mumbai, IN · Dec 2020 – Aug 2021

- Implemented 80+ order execution algorithms, from time and price triggers to market making, streaming ~10,000 orders a day over WebSockets at under 30 ms latency for funds up to ₹200 crore (~$24M).
- Built an Angular and Express risk dashboard on MongoDB that cut the time to assess exposure and place orders by 50%.
- Deployed the platform on AWS in the region closest to the broker, making order execution 20% faster.

**Accio.ai** *Software Engineer Intern* Mumbai, IN · Apr – Sep 2018

- Built the analytics ingestion pipeline for ad clicks, impressions and views (~60M events a month) into MongoDB, plus the REST API for ad payment reporting.

## Education

**Master of Science in Computer Science** *University of Massachusetts, Boston* CGPA 3.7/4 · 2021 – 2023

- Graduate Teaching Fellow for CS240 Programming in C, two semesters, classes of 51 and 67 students.

## Projects & Open Source

**Option Chain Analysis** · Live NSE Nifty option chain, every strike classified by price and open interest change · Node.js — 100+ stars · 2019

**Broker SDKs** · Kite Connect (Go), XTS v2 and 5paisa (Node.js), Hummingbot client (Python), option pricing formulas — 2020 – 2025

**UMB Trading App** · Options trading interface designed for beginners · Figma, JavaScript — 2021

## Skills

| | |
|---|---|
| **Languages** | Go, Python, TypeScript, JavaScript, C++, C, SQL |
| **Trading** | Options Greeks, backtesting, order execution, market making, MT5/MQL5 EAs, WebSockets, NSE option chain data, broker APIs |
| **Data & infra** | AWS, CloudFormation, Secrets Manager, GitHub Actions, Kubernetes, Docker, Snowflake, Airflow, Kafka, Spark, Redis, MongoDB, Linux |